MIKE WONG

mikedwong@cs.princeton.edu · michaeldwong.github.io · Google Scholar

SUMMARY

Fifth-year CS PhD candidate at Princeton University building efficient, scalable machine learning systems for real-world deployment. Research sits at the intersection of computer networks, systems, and machine learning, using domain-specific insights to filter, compress, and prioritize data before expensive inference stages. Track record of designing application-aware data processing pipelines that improve performance, cost, and latency across web agents, vision AI, and network security. Published at top venues (NSDI, SIGCOMM, CoNEXT). Graduating late 2026.

EDUCATION

Princeton University *Aug 2021 – Late 2026 (expected)*

Ph.D. in Computer Science. Advisor: Ravi Netravali. M.A. conferred Sep 2023.

New York University *Aug 2015 – Dec 2019*

B.A. in Computer Science, minor in Mathematics. Graduated with Honors.

EXPERIENCE

Graduate Research Assistant, Princeton University *Aug 2021 – Present*

Advisor: Ravi Netravali

- Built **MadEye** (NSDI '24, first author), a camera-server system that dynamically filters and prioritizes spatial video data, combining knowledge distillation with a fast spatial search algorithm to cut live inference streaming costs by **2–3.7×** while matching or improving baseline tracking accuracy.
- Designed and built **Glenfinnan** (SIGCOMM NAIC '26, first author), a hardware-accelerated processing engine that pushes vision AI pipeline orchestration into programmable network fabric via SmartNICs, eliminating CPU-bound deserialization and orchestration bottlenecks.
- Co-authored a detection-guided attention steering method for vision-language models (ICML SCALE '26) that masks task-irrelevant visual regions before transformer cross-attention, accelerating spatial model context processing.

Research Intern, Cloud Systems Reliability Group, Microsoft Research *May – Aug 2025*

- Developed **Skim** (arXiv '26, first author), a speculative execution stack for LLM web agents that learns reusable website layouts to predict data acquisition strategies before runtime.
- Cut web agent per-task inference costs by **1.9×** and task latency by **33.4%** on production web automation benchmarks while preserving target accuracy.

Research Intern, Networking Research Group, Microsoft Research *Jun – Aug 2022*

- Developed **NetVigil** (NSDI '24, co-first author), a low-cost east-west anomaly detector replacing heavy packet traces with lightweight flow summaries, cutting telemetry overhead by **2.7–16.7×**.
- Evaluated on a production-scale 16-VM cluster using graph neural networks and contrastive learning, beating SOTA baseline AUCs by 0.22 on average; led to a filed US patent and open-source benchmark (Yatesbury).

Computer Scientist, National Security Agency *Feb 2020 – Aug 2021*

- Maintained and contributed core features to **Ghidra**, the agency's open-source software reverse-engineering framework used across global cybersecurity research and defense.
- Developed **Marvolo** (ECML PKDD '23, first author), an automated data augmentation engine applying semantics-preserving binary filters, achieving up to 10% ML detection AUC improvement with a **79×** rewrite speedup over naive binary writing.

Research Assistant, Rutgers University / NYU Courant *May 2019 – Jan 2021*

Advisors: Srinivas Narayana, Anirudh Sivaraman

- Co-developed **K2** (SIGCOMM '21), a stochastic program-synthesis eBPF compiler reducing kernel extension sizes by **6–26%** and accelerating equivalence checking by 5 orders of magnitude.
- Developed **Druzhba** (CoNEXT '20, first author), a high-fidelity programmable-switch hardware simulator, and co-developed **Chipmunk** (SIGCOMM '20), a synthesis-based switch compiler yielding a **51×** average compilation speedup.

PUBLICATIONS

* denotes equal contribution. Full list at michaeldwong.github.io.

1. **Skim: Speculative Execution for Fast and Efficient Web Agents.** Mike Wong, Kevin Hsieh, Suman Nath, Ravi Netravali. arXiv:2605.16565, 2026.
2. **Glenfinnan: SmartNIC-Accelerated Data Processing for Efficient Vision AI Pipelines.** Mike Wong, Ulysses Butler, Emma Farkash, Praveen Tammana, Anirudh Sivaraman, Ravi Netravali. ACM SIGCOMM NAIC 2026.
3. **Detection-guided Attention Steering for Vision Language Models.** Alan Zhang, Rui Pan, Mike Wong, Ravi Netravali. ICML SCALE 2026.
4. **MadEye: Boosting Live Video Analytics Accuracy with Adaptive Camera Configurations.** Mike Wong, Murali Ramanujam, Guha Balakrishnan, Ravi Netravali. USENIX NSDI 2024.
5. **NetVigil: Robust and Low-Cost Anomaly Detection for East-West Data Center Security.** Kevin Hsieh*, Mike Wong*, Santiago Segarra, Sathiya Kumaran Mani, Trevor Eberl, Anatoliy Panasyuk, Ravi Netravali, Ranveer Chandra, Srikanth Kandula (*equal contribution). USENIX NSDI 2024.
6. **Marvolo: Programmatic Data Augmentation for Deep Malware Detection.** Mike Wong, Edward Raff, James Holt, Ravi Netravali. ECML PKDD 2023. *Early version at AI4Cyber/MLHat Workshop, ACM KDD 2022.*
7. **Synthesizing Safe and Efficient Kernel Extensions for Packet Processing.** Qiongwen Xu, Michael D. Wong, Tanvi Wagle, Srinivas Narayana, Anirudh Sivaraman. ACM SIGCOMM 2021. *Also at BPF & Networking Summit, LPC 2021.*
8. **Testing Compilers for Programmable Switches Through Switch Hardware Simulation.** Michael D. Wong, Aatish Kishan Varma, Anirudh Sivaraman. ACM CoNEXT 2020.
9. **Switch Code Generation Using Program Synthesis.** Xiangyu Gao, Taegyun Kim, Michael D. Wong, Divya Raghunathan, Aatish Kishan Varma, Pravein Govindan Kannan, Anirudh Sivaraman, Srinivas Narayana, Aarti Gupta. ACM SIGCOMM 2020.

TECHNICAL SKILLS

Languages	Python, C++, C, Rust, Java, JavaScript, Assembly
ML / Systems	Distributed model serving, infrastructure optimization, computer vision, SmartNIC orchestration
LLMs / Agents	Agent speculative execution, knowledge base ETL pipelines, ReAct, prompt engineering, vLLM
Networking	eBPF, Verilog, Linux kernel programming, P4, programmable switches, packet processing
Tools	Ghidra, Git, Docker, Kubernetes, Linux Sys Admin, LaTeX

PATENTS

Detecting Network Anomalies Using Network Flow Data. Tsuwang Hsieh, Santiago Martin Segarra, Sathiya Kumaran Mani, Srikanth Kandula, Michael Dean Wong. US patent App. 18/367,775.

TEACHING & VOLUNTEERING

Instructor, Prison Teaching Initiative (Princeton University) *Spring 2025 – Present*
Volunteer instructor teaching mathematics (MATH 020, MATH 101) to incarcerated students in NJ state correctional facilities.

Teaching Assistant, Princeton University Dept. of Computer Science *Spring – Fall 2023*
Assisted instruction, built grading infrastructure, held office hours for **COS 316** (Principles of Computer System Design) and **COS 561** (Advanced Computer Networks).